

# A1

## Information Presentation Response

1. **What is something you learned from the presentation that you didn't know before?**
  - I learned that ACM Code of Ethics including the principles of enhancing public awareness and understanding of computing technologies and their social influence highlights an important and proactive role for professionals in public education.
2. **What area would you like to see explored in more detail in a follow-on presentation?**
  - I want to explore more on how IEEE and ACM code address emerging technologies, such as AI and blockchain, especially under the background of ethical dilemmas and practical case studies.

## Ethical Advice

### Advice for Sandra

- Since Sandra is the president of Med-E-Clinic Systems, she faces both ethical and business dilemma about whether to work with an abortion clinic. It will raise moral and reputational concerns. While the ACM and IEEE Codes of Ethics emphasize fairness, integrity, and prioritizing public welfare (ACM, 2018; IEEE, 2020), Sandra must also consider the potential backlash from stakeholders, as abortion remains a controversial issue and homeopathy lacks scientific credibility (Reynolds-Wright et al., 2021; Smith, 2020).
- To address these challenges, Sandra should establish clear corporate policies on client selection and ethical considerations, ensuring fairness and transparency to prevent accusations of bias (Ferrell & Fraedrich, 2021). She must also improve conflict resolution strategies to manage workplace tensions, ensuring professional behavior among employees (Jehn & Mannix, 2022). Additionally, Sandra must reject unethical conduct, such as Sam's suggestion to install faulty software, as it would violate professional standards and expose the company to legal liability (Solove, 2021).
- Finally, Sandra must balance ethics, financial stability, and employee concerns by upholding professional integrity, fostering open communication, and making well-structured decisions. By doing so, she can maintain the company's credibility and lead with ethical responsibility.

### Advice for Sam

- Sam faces an ethical dilemma, torn between his personal beliefs and professional responsibilities. While he strongly opposes abortion, his suggestion to sabotage the abortion clinic's software is unethical, illegal, and a violation of professional standards (ACM, 2018; IEEE, 2020). Sabotaging a client would not only harm Med-E-Clinic Systems' reputation but also expose the company to legal consequences (Solove, 2021).
- Rather than engaging in unethical behavior, Sam should professionally express his concerns and request a reassignment if necessary (Ferrell & Fraedrich, 2021). Additionally, his personal attack on Cheryl was unprofessional, highlighting the need for workplace conflict resolution strategies (Jehn & Mannix, 2022). Finally, Sam must balance his moral beliefs with his professional duties, ensuring that he upholds ethical integrity, workplace professionalism, and respect for colleagues and clients.

## Advice for Cheryl

- Cheryl faces a professional dilemma, as she opposes working with the homeopathic clinic due to her belief that it is unscientific and potentially harmful. However, her willingness to sabotage the clinic undermines her ethical integrity and professional credibility (ACM, 2018; IEEE, 2020). Engaging in unethical behavior would damage Med-E-Clinic Systems' reputation and expose the company to legal risks (Solove, 2021).
- Rather than taking unethical action, Cheryl should express her concerns professionally and request reassignment if necessary (Ferrell & Fraedrich, 2021). She must also recognize that her role is to provide quality software solutions, not to endorse medical practices. By focusing on her professional responsibilities while maintaining open communication, Cheryl can navigate this conflict ethically and uphold both her values and professional integrity.

## Report on a Software Disaster

### Telangana exam result suicides

#### Introduction

- The Telangana exam result suicides of 2019 highlight the devastating real-world consequences of software failures and poor oversight. Over 26 students in Telangana, India, took their own lives after the Telangana State Board of Intermediate Education (TSBIE) mistakenly reported their exam results as failing due to technical errors (BBC News, 2019). These errors were the result of faulty software algorithms, inadequate testing, and a lack of validation protocols in the exam processing system. The tragedy serves as a wake-up call for policymakers, educational institutions, and software developers to prioritize software quality assurance, mental health awareness, and accountability in high-stakes situations.

## The Disaster

- The Telangana exam result suicides were caused by a faulty software system that erroneously marked passing students as failed. The private IT firm responsible for grading the results used flawed algorithms and improper data handling, which led to widespread errors in student scores (The Hindu, 2019). Additionally, TSBIE failed to implement validation mechanisms or conduct audits, allowing the errors to go undetected before publication (BBC News, 2019). When students received incorrect failing grades, they experienced extreme emotional distress, as academic success is often directly tied to career opportunities and social standing in India (Hindustan Times, 2019). Beyond technical failures, this incident underscores the ethical responsibilities of software developers and education boards. If proper quality assurance, validation protocols, and student mental health programs had been in place, these tragic deaths could have been prevented (IEEE, 2020).

## Key Issues

- Firstly, the software development process lacked proper oversight. Secondly, the lack of contingency plans or data audits meant that the errors were not only overlooked but also left uncorrected for an extended period (Hindustan Times, 2019). Thirdly, mental health support for students was severely lacking.

## Lessons Learned

- First, software used for high-stakes applications must undergo rigorous quality control and independent verification. Secondly, validation and auditing mechanisms must be mandatory in all exam-processing systems. Results should undergo multiple verification steps before being released, and automated error detection mechanisms should be implemented to prevent similar mistakes (BBC News, 2019). Third, mental health support must be integrated into educational institutions. Finally, crisis management protocols need to be in place for handling such emergencies. Institutions should establish rapid-response teams to address errors immediately, communicate transparently with students and families, and take swift corrective actions to prevent unnecessary panic and distress (Hindustan Times, 2019).

## Conclusion

- The Telangana exam result suicides serve as a stark reminder of the ethical, technical, and psychological risks associated with software errors in high-stakes situations. This incident was not just a technological failure but a failure in governance, quality control, and student mental health support. To prevent future tragedies, governments, educational

boards, and software developers must prioritize software accuracy, transparency, and student well-being. By enforcing strict quality control measures, providing adequate mental health resources, and ensuring immediate crisis response, policymakers and institutions can help prevent such devastating outcomes. The Telangana case should serve as a lesson in responsibility, reinforcing the need for technology to serve as a tool for empowerment rather than destruction.

# Reference

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